REAL-TIME 3D HEAD POSE ESTIMATION USING BOTH GEOMETRY AND LEARNING

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ABSTRACT

In this paper we propose a very efficient and low-cost method for real-time 3D object pose estimation, suitable for operation under severe changes in the illumination conditions. The method consists of a geometry-based and learning-based stage: first a geometry-based pose estimation algorithm is used to obtain pose parameter values, which are utilized as supervised information during a following learning-based stage. The efficacy of the proposed approach is illustrated on a real-world application in a head pose based facial-gesture interface for automatic wheelchair navigation.

Index Terms— 3D Head Pose Estimation, Facial Gesture Interface, Combinatorial Optimization, Stereo Images

1. INTRODUCTION

3D pose estimation is an important problem in computer vision, with many potential applications in object tracking and recognition, virtual and augmented reality, human-computer interfaces, robotic vision, etc. Many different approaches have been proposed to solve this problem, but in this paper we will focus our attention on the learning-based approach [1], used in the context of 3D head pose estimation. Learning-based pose estimation methods (e.g. [2]) generally learn a map between a set of training samples and pose-space. Important advantages of learning based methods over purely geometric methods [3] include robustness under real-world conditions, speed, domain-independence, and so on. However, a major problem of this approach is the need to collect a large set of training samples labeled with the corresponding pose parameter values. Providing this supervised information is costly and very often necessitates the use of specialized (and expensive) sensors which can be an unacceptable solution for many applications.

In this paper we address these issues by proposing a new two-staged approach, which combines a geometry-based pose estimation method (the first stage) for obtaining pose parameter values, to be used as supervised information during the following learning-based stage (the second stage). The geometry-based pose estimation method recovers the 3D pose of the object of interest from a small number of detected landmarks.

However, detecting the landmarks on the object under challenging illumination conditions is a difficult problem itself, and performance can be unreliable under real-world conditions. Therefore, we propose to perform this initial pose estimation stage under relatively stable illumination conditions and use the obtained pose labels as supervised information for the training samples during the second learning stage. The important point here however is that these pose labels are coupled not with the usual texture (intensity) images, but with disparity images obtained from a stereo camera. The disparity images contain information about the 3D shape of the objects of interest and are much more stable under severe illumination changes than the texture images.

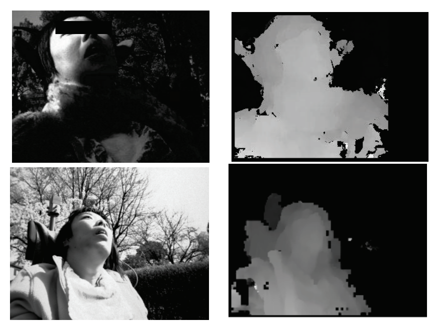


Fig. 1. Disparity images (right) are much more stable under illumination changes than texture images (left)

Therefore, the disparity images taken under controlled illumination conditions during the first stage (*pose labels acquisition*), and the disparity images obtained under test (*operation*) conditions would not differ very much from each other (see Fig.1 for an example). Thus, during the second (*learning*) stage, a map is learned between the set of disparity images (the training samples) and their corresponding pose-space parameters (the supervised information, i.e. pose labels obtained during the first, geometry-based stage – *pose labels acquisition*).

2. METHODS

We illustrate the proposed method using a concrete real-world application – a facial gesture interface controlling the autonomous navigation of a wheelchair for people suffering from cerebral palsy, who are paralyzed below the neck and cannot control their wheelchair using the limbs or even speech. In this case, the 3D pose of the face needs to be estimated, and the continuously estimated pan and tilt angles of the head are used to control an elaborate menu which directs the motion of the wheelchair. Wheelchair command operations like "forward", "right turn", "left turn", "turn on the spot", "reverse", "change speed", "stop", etc., are defined based on the results of clinical trials (see [4] for further details on the design of the head gesture interface).

During the first stage of the proposed method (see Fig. 2), a small number of markers (used for ease of detection) are attached to the face, and a sequence of both texture and disparity images of the moving face are obtained. The markers are detected in the texture images based on their color and shape properties, and reconstructed in 3D by triangulation. Then 3D pose (pan, tilt, roll, X, Y, Z) is obtained using the PC-SCO algorithm explained briefly below, and used as a pose label to the corresponding disparity image.

At the same time, features are extracted from the disparity images for learning. For features we used a modified 3D version of the shape context descriptor [5], i.e. a 3D histogram was constructed from each disparity image, where the first two dimensions of the histogram bins corresponded to a log-polar map of the disparity X and Y coordinates, and the third dimension of the histogram bins the disparity values (that is, corresponds to distance Z). The map between pose space and the training samples was learned using a nonlinear manifold learning method [6]. After learning is complete, the pose of any new face sample is estimated automatically by mapping it to pose space using the learned map, i.e. neither markers nor any other supervised information is required at test time. Also, since the disparity images provide only rough shape information (it is not possible to discriminate between different subjects based on this information), the proposed method is generally user-independent.

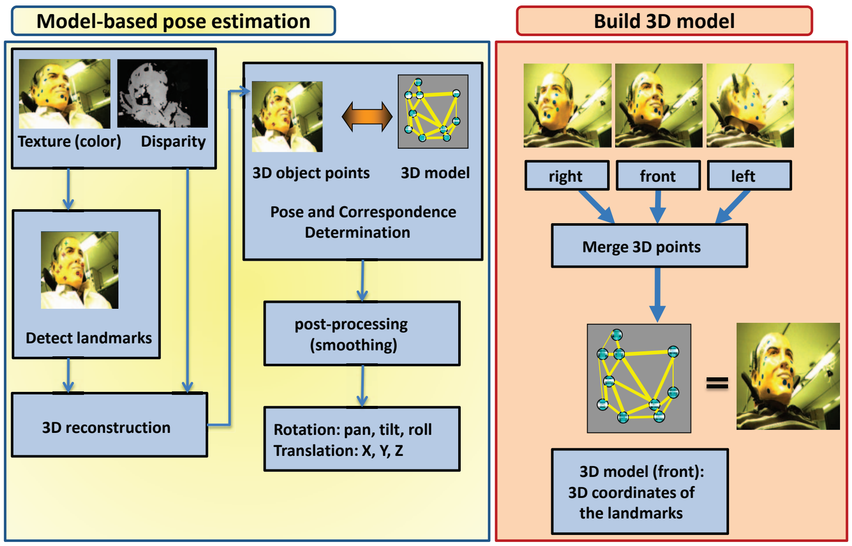


Fig. 2. The model-based pose estimation algorithm

Below, we will describe in some detail our pose estimation algorithm, PC-SCO (Pose and Correspondence through Stochastic Combinatorial Optimization). Further technical details are in [7]. In *PC-SCO*, the point correspondence problem is formulated as a linear assignment problem, where the assignment cost is optimized through a stochastic search through correspondence space. We consider the minimization of the following cost function:

$$E(\mathbf{M},\mathbf{R},\mathbf{t}) = \min_{\mathbf{M},\mathbf{R},\mathbf{t}} \sum_{j=1}^{M}\sum_{i=1}^{N} m_{ij}\left(\left\|y_j - \mathbf{R}x_i - \mathbf{t}\right\|^2 - \alpha\right) \tag{1}$$

where X and Y are correspondingly the test-sample and 3D model's points, ($\mathbf{R}$, $\mathbf{t}$) are the sought pose parameters (rotation matrix and translation), and $\mathbf{M}$ is a binary correspondence matrix, i.e. m_{ij} = 1 if point X_i in the object-view corresponds to point Y_j in the 3D model prepared in advance (Fig. 2, right), and m_{ij} = 0 otherwise. There should be at most one '1' in each row of $\mathbf{M}$, since the correspondence between X and Y is one-to-one. If there is not a single '1' in the i-th row, this means that X_i is considered an outlier.

To solve the combinatorial optimization problem (1), correspondence space is stochastically sampled by a Markov process, and new solutions are accepted or rejected according to the Metropolis algorithm with the following transition probabilities:

$$P(\mathbf{M}^{old} \to \mathbf{M}^{new}) = \begin{cases} 1 & \textit{if } \Delta E(\mathbf{M}) \le 0 \\ \exp(-\Delta E(\mathbf{M})/T) & \textit{otherwise} \end{cases} \tag{2}$$

where $\Delta E(\mathbf{M}) \equiv E(\mathbf{M}^{new}) - E(\mathbf{M}^{old})$. State changes corresponding to decreases in the cost function are always accepted, while in addition, to avoid local minima, state changes corresponding to an increased cost are accepted with an exponentially weighted probability, determined by the temperature parameter T. The temperature is gradually reduced during the stochastic search process according to an annealing schedule $T(k+1) = rT(k)$, where the *annealing rate* r was set to 0.95.

Another important detail is to determine the way in which the configuration $\mathbf{M}^{old}$ has to be perturbed in order to obtain the new configuration $\mathbf{M}^{new}$, or the definition of a "move" between two configurations. Each move randomly selects some m_{ij} whose binary value is flipped as

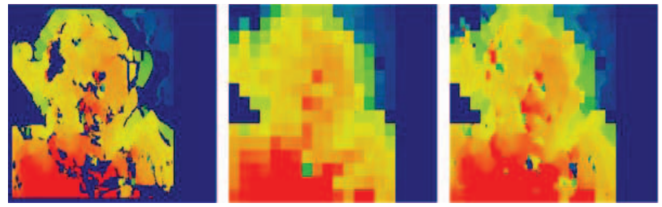
$$m_{ij}^{new} := \begin{cases} 1 & if \;\; m_{ij}^{old} = 0 \\ 0 & if \;\; m_{ij}^{old} = 1 \end{cases} . \tag{3}$$

If $m_{ij}^{new} = 1$, i.e. X_i is considered as a prospective match to Y_j, all m_{ik}^{new} ($k \neq j$) also have to be set to '0', to ensure that the i-th row sums to 1. Alternatively, if $m_{ij}^{new} = 0$, , i.e. X_i does not correspond any more to Y_j, it needs to be checked that at least 3 rows in $\mathbf{M}$ contain a '1', and if this is not the case, to set $m_{ik}^{new} = 1$for some randomly chosen k, to ensure that we have at least 3 correspondences, the minimum needed to be able to determine the pose.

PC-SCO starts with a high temperature value $T(0)$ and a randomly chosen state configuration $\mathbf{M}$, for which ($\mathbf{R}$, $\mathbf{t}$) are calculated with Horn's quaternion method [8]. Then $\mathbf{M}$ is randomly perturbed as explained above, and if the new correspondence is accepted, which depends on (2), pose is recalculated for the new configuration. This process is repeated a sufficient number of times, before T is lowered according to the annealing schedule, and this continues until T reaches a sufficiently low value. At the end, $\mathbf{M}$ corresponding to the lowest $E(\mathbf{M})$ is used to calculate the final ($\mathbf{R}$, $\mathbf{t}$). Because only a small number of landmarks (less than twenty) are sufficient to achieve reliable pose estimation, the above combinatorial optimization problem is solved very fast, allowing real-time performance to be achieved.

3. EXPERIMENTS

In order to assess the plausibility of the proposed approach, we performed the following experiments. 11 blue markers were attached to the face of a subject, and both color texture images (640x480 pixels) and disparity images (320x240 pixels) were obtained by a stereo camera at 30 fps, as the subject was moving his face while sitting in a wheelchair. The video sequence contained 22645 frames in total. Based on clinical requirements, the camera was fixed on the wheelchair at 20 deg to the right of the user, and the user was sitting in a way which closely simulated the posture of actual users suffering from cerebral palsy. Such users cannot hold an upright position and lean backwards to some extent, as a result of which the head is tilted about 30-40 deg upwards.



(a) (b) (c)

Fig. 3. (a) disparity image of size 320x240 pixels, with multiple "holes" at which the stereo matching process has failed; (b) image representing the lowest level (20x15 pixels) of the scale pyramid made from the original disparity image in (a); (c) the image in (a) with "holes" filled.

These requirements greatly increase the difficulty of the head pose estimation process (compared to the case of a frontally located camera and a user sitting in an upright position). Areas in the disparity images containing "holes" due to local failures of the stereo matching process were interpolated from the neighboring areas by using an interpolation on a scale pyramid prepared from the original image (refer to Fig. 3 for an example). Missing data at each level of the pyramid was interpolated from its immediate neighbors at the same level, and if this was not possible, then from its corresponding neighbors at a lower level. In the resulting disparity images, since the distance to the head area is approximately known, first the background is removed by simple thresholding. Then the head area is easily segmented from the shoulders area by utilizing the large shape discontinuity existing between them. A simple 3D model of the face was obtained in advance, as shown on the right side of Fig. 2: the 3D coordinates of the markers detected in a left, frontal and right view of a face are recovered and merged in a common coordinate frame. Then this model is matched using PC-SCO to the recovered 3D coordinates of the markers detected in each frame. The obtained head pose is used as supervised information and training performed as described in section 2. For evaluation of the performance of pose estimation, we gathered an alternative set of pose labels for the same 22645 frames sequence, obtained by using a (much more expensive) Polhemus magnetic sensor, attached to the head of the subject. Then the whole video sequence was split into two, and the first half was used for training and the second half for test. This was then repeated while using the second half for training and the first half for evaluation, i.e. the whole 22645 frames sequence was used for the evaluation. As we wanted to show whether the proposed geometry-based method (using the PC-SCO algorithm) for pose estimation can achieve a similar performance to a magnetic sensor based pose estimation, we alternatively used these two methods during the first stage (pose labels acquisition). Therefore, in this experiment we compare the performance of the system when the supervised information for the learning was provided by (a) using PC-SCO and (b) using a magnetic sensor. Fig. 4 shows comparison results between the proposed method and the magnetic sensor on a typical subsequence. In Fig. 4, ground truth obtained from a magnetic sensor (shown in blue) is compared to estimation results obtained when (a) learning used pose labels obtained with markers and using the PC-SCO algorithm (yellow graph), and (b) learning used pose labels obtained from the magnetic sensor (graph in magenta). As can be seen from Fig. 4 both pan and tilt (which are used to control the facial interface) are estimated with a satisfactory precision for the needs of the gesture interface. The mean and standard deviation of the absolute pose estimation error for the whole 22645 frames sequence are compared in Fig. 5.

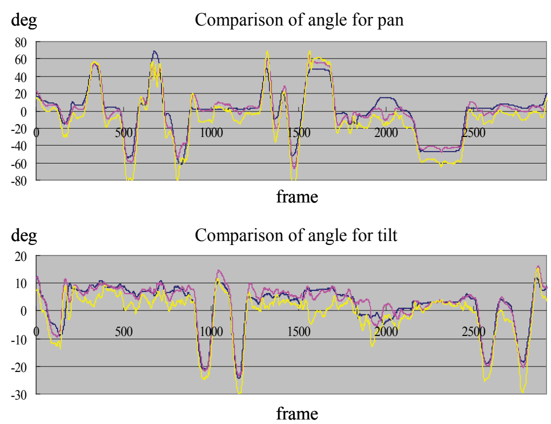


Fig. 4. Comparison between learning-based pose estimation when using pose label data obtained by PC-SCO (*yellow*) or using the Polhemus magnetic sensor (*magenta*). Ground truth is shown in *blue*.

Additionally, another experiment was conducted to determine the recognition rate for the 3D face-pose based face gesture interface controlling the wheelchair. Experimental data was collected over 3 days (86 minutes in total) and contained 1493 instances of 4 predefined basic face gestures (front, right, left, down). For the whole sequence, we had only 27 instances of erroneous gestures, which is a 98.19% recognition rate (see Table 1).

4. CONCLUSION

In this paper we have proposes a very efficient and low-cost method for real-time 3D object pose estimation, suitable for operation under severe changes in the illumination conditions. Our method combines a geometry-based pose estimation algorithm for obtaining pose parameter values, to be used as supervised information during a following learning-based stage. The efficacy of the proposed approach was also illustrated on a real-world application in a head pose based facial-gesture interface for automatic wheelchair navigation.

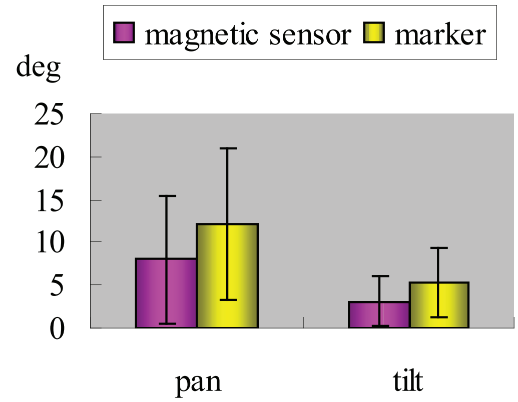


Fig. 5. Mean and standard deviation for absolute pan and tilt estimation error.

Recognition rates [%]	**total**	front	right	left	down
	98.19	97.84	98.45	98.24	99.09

Table 1. Recognition rates (in percents) for the 3D face-pose based face gesture interface when the pose labels acquisition was done using the PC-SCO algorithm.

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